Preliminaries of Machine Learning

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Chapter 1

Introduction

This is some notes about preliminaries of machine learning, based on the book by Kevin Murphy.

1.1 Notation

$\mathbb{E}_{x \sim p(x)}$ means that the expectation is taken about x where x has distribution $p(x)$.

Chapter 2

Probability

2.1 Basic Definitions

Definition 2.1.0.1 (Probability Space)

A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where

- Ω is a non-empty set called the *sample space*;
- $\mathcal{F}$ is a σ-algebra of subsets of the sample space Ω;
- $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a probability measure.

2.2 Lists of Probability Distributions

2.2.1 Discrete Distributions

Name	Support	Parameters	PMF	Mean, Variance
Bernoulli	$\{0, 1\}$	$p \in \{0, 1\}$	$\mathrm{Ber}(x\|p) = p^x(1-p)^{1-x}$	$\mathbb{E}[X] = p, \mathbb{V}[X] = p(1-p)$
Binomial	$\{0, 1, 2, \ldots, N\}$	$N, p \in [0, 1]$	$\mathrm{Bin}(x\|N, p) = \binom{N}{x} p^x (1-p)^{N-x}$	$\mathbb{E}[X] = Np, \mathbb{V}[X] = Np(1-p)$
Categorical	$\{0, 1, 2, \ldots, N\}$	$\boldsymbol{\theta}$	$\mathrm{Cat}(x\|\boldsymbol{\theta}) = \prod_{k=1}^{K} \theta_k^{\mathbb{I}(x=k)}$	
Multinomial	$\boldsymbol{x}$ where $\sum_{k=1}^{K} x_k = N$	$N, \boldsymbol{\theta}$	$\mathrm{Mult}(\boldsymbol{x}\|\boldsymbol{\theta}) = \binom{N}{x_1, x_2, \ldots, x_K} \prod_{k=1}^{K} \theta_k^{x_k}$	
Poisson	$\{0, 1, 2, \ldots\}$	$\lambda > 0$	$\mathrm{Pois}(x\|\lambda) = e^{-\lambda} \dfrac{\lambda^x}{x!}$	$\mathbb{E}[X] = \lambda, \mathbb{V}[X] = \lambda$

2.2.2 Continuous Distributions

Name	Support	Parameters	PDF	Mean, Variance
Gaussian	$\mathbb{R}$	μ, σ	$\mathcal{N}(\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$	$\mathbb{E}[X] = \mu, \mathbb{V}[X] = \sigma^2$
Student t-distribution	$\mathbb{R}$	μ, σ	$\mathcal{T}_\nu(x\|\mu, \sigma^2) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$	$\mathbb{E}[X] = 0$ for $\nu > 0$, $\mathbb{V}[X] = \frac{\nu}{\nu-2}$ for $\nu > 2$
Cauchy	$\mathbb{R}$	μ, λ	$\mathcal{C}(x\|\mu, \lambda) = \frac{1}{\lambda\pi(\frac{x-\mu}{\lambda})^2}$	undefined
Laplace	$\mathbb{R}$	μ, b	$\text{Laplace}(x\|\mu, b) = \frac{1}{2b} \exp\left(-\frac{\|x-\mu\|}{b}\right)$	$\mathbb{E}[X] = \mu, \mathbb{V}[X] = 2b^2$
Gamma	$\mathbb{R}^+$	$a, b > 0$	$\text{Ga}(x\|\text{shape} = a, \text{scale} = b) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-xb}$	$\mathbb{E}[X] = ab, \mathbb{V}[X] = ab^2$
Exponential	$\mathbb{R}^+$	λ	$\text{Expon}(x\|\lambda) = \lambda e^{-\lambda x}$	$\mathbb{E}[X] = \lambda, \mathbb{V}[X] = \lambda^2$
Chi-squared	$\mathbb{R}^+$	none	$\chi^2_\nu(x) = \text{Ga}(x\|\text{shape} = \nu/2, \text{scale} = 2)$	$\mathbb{E}[X] = \nu, \mathbb{V}[X] = 2\nu$

Chapter 3

Statistics

This part is about guessing the parameters of our probability models shown in the previous section. Most of the estimation will take the form of $\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}}\ \mathcal{L}(\boldsymbol{\theta})$, where the $\boldsymbol{\theta}$ is our parameter to be estimated, and the $\mathcal{L}$ is the *loss function*.

There will be two main approaches, namely **frequentist statistics** and **Bayesian statistics**. Frequentist method treats parameters $\boldsymbol{\theta}$ as fixed value, and the data $\mathcal{D}$ as unknown and changes. Conversely, Bayesian method treats parameters $\boldsymbol{\theta}$ as unknown, and the data $\mathcal{D}$ as fixed and known.

However, note that the model may not take any parameters; such models can be adjusted by using **nonparametric methods**.

3.1 Bayesian Statistics

Bayesian statistics treats parameters $\boldsymbol{\theta}$ as unknown, and the data $\mathcal{D}$ as fixed and known. We use Bayes rule to update our uncertainty about parameters when the data is given;

$$p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\boldsymbol{\theta})p(\mathcal{D}|\boldsymbol{\theta})}{p(\mathcal{D})} = \frac{p(\boldsymbol{\theta})p(\mathcal{D}|\boldsymbol{\theta})}{\int p(\boldsymbol{\theta}')p(\mathcal{D}|\boldsymbol{\theta}')d\boldsymbol{\theta}'}$$

Here, $p(\boldsymbol{\theta})$ is called the **prior**, and $p(\mathcal{D}|\boldsymbol{\theta})$ is called the **likelihood**. $p(\boldsymbol{\theta}|\mathcal{D})$ is called the **posterior**, and $p(\mathcal{D})$ is called the **marginal likelihood**.

After the calculation of posterior distribution, we can try to predict the future observations by calculating the **posterior predictive distribution**:

$$p(x|\mathcal{D}) = \int p(x|\boldsymbol{\theta})p(\boldsymbol{\theta}|\mathcal{D})d\boldsymbol{\theta}$$

We are marginalizing out all the unknown parameters. If the calculation is too expensive, we can just plug-in the paramter value we estimated, like MLE or MAP.

$$p(x|\mathcal{D}) \approx p(x|\hat{\boldsymbol{\theta}})$$

This is called **plug-in approximation**.

3.1.1 Selecting the Prior

Selecting appropriate prior for parameters are one of the challenging problem of Bayesian method. There are several choices;

- conjugate prior
- uninformative prior

- hierarchical prior
- empirical prior

3.2 Frequentist Statistics

Frequentist statistics assumes that there is a true parameter value $\boldsymbol{\theta}$ and tries to estimate the parameter value by changing the data. The change of data will change the quantity which is used to estimate the parameter, and we use this change to check which is the best estimation.

Even if the frequentist method has some drawbacks and even paradoxes, several concepts such as cross validation is useful even in the Bayesian method. Here, the basic concepts are shown.

3.2.1 Sampling Distribution

In frequentist statistics, uncertainty is represented by the **sampling distribution** of a estimator. Here, estimator can be MLE, MAP estimate or method of moments estimate.

Definition 3.2.1.1 (Sampling Distribution)

Let $\boldsymbol{\Theta}(\mathcal{D})$ be an estimator of parameter $\boldsymbol{\theta}$. Then, the sampling distribution of $\boldsymbol{\Theta}(\mathcal{D})$ is the distribution obtained by simply keep changing the data set, $\mathcal{D}$. More precisely, let $\tilde{\mathcal{D}}^{(s)}$ be an s-th sample from true data distribution. Then,

$$p(\boldsymbol{\Theta}(\tilde{\mathcal{D}}) = \boldsymbol{\theta}) \approx \frac{1}{S}\sum_{s=1}^{S} \delta(\boldsymbol{\theta} - \boldsymbol{\Theta}(\tilde{\mathcal{D}}^{(s)}))$$

where S is size of the sample.

3.2.1.1 Bootstrap Approximation of Sampling Distribution

Since the true distribution of data set is unknown, we can't easily get the sampling distribution of estimator. Instead, we can use **bootstrap** method.

Definition 3.2.1.2 (Bootstrap Approximation)

Let $\mathcal{D}$ be the given data set, and let $p(x|\boldsymbol{\theta}^*)$ be the true distribution of data set.

- parametric bootstrap: plug-in the estimator value in proability distribution and create sampling distribution. More precisely, use $\tilde{\mathcal{D}}^{(s)} = \{\boldsymbol{x}_n \sim p(x|\boldsymbol{\Theta}(\mathcal{D}))|n = 1, 2, \ldots, N\}$ instead of $\{\boldsymbol{x}_n \sim p(x|\boldsymbol{\theta}^*)|n = 1, 2, \ldots, N\}$.
- nonparametric bootstrap: sample datapoints from the original dataset.

3.2.1.2 Bootstrap vs Bayesian

An estimator obtained by bootstrap method and posterior distribution are quite different. But, in certain condition, where estimator is MLE and the prior for posterior distribution is not so strong, they can be quite similar. Check this theorem.

Bootstrap method can actually be slower than posterior sampling, since bootstrap method requires some number of sampling, but posterior need only one sampling.

3.2.1.3 Asymptotic Normality of Sampling Distribution

How can one ensure that the sampling distribution is really useful? i.e., can we really approximate real parameter value with sampling distribution? The answer is yes, under certain conditions, when the estimator is MLE.

Theorem 3.2.1.1 (Asymptotic Normality of Sampling Distribution)

Under various technical conditions, we have

$$\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*) \to \mathcal{N}(\mathbf{0}, \mathbf{F}(\boldsymbol{\theta}^*)^{-1})$$

where $\mathbf{F}(\boldsymbol{\theta}^*)$ is **Fisher information matrix**, and $\boldsymbol{\theta}^*$ is the true parameter value.

Since the Fisher information matrix equals the expectation of Hessian of log likelihood, which is shown here, and Hessian of likelihood measure the amount of curvature of the surface of log likelihood, we can infer that the problem with peaked likelihood will give good estimate of parameter when using sampling distribution.

3.2.2 Fisher Information Matrix

The Fisher information matrix plays a key role in frequentist statistics, and it is also used in Bayesian statistics (Jeffery's prior), and optimization.

Definition 3.2.2.1 (Score Function)

The score function is defined to be the gradient of the log likelihood:

$$\boldsymbol{s}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \log p(\boldsymbol{x}|\boldsymbol{\theta})$$

Definition 3.2.2.2 (Fisher Information Matrix)

Let $\boldsymbol{s}(\boldsymbol{\theta})$ be the score function. The Fisher information matrix is defined to be the covariance of the score function:

$$\mathbf{F}(\boldsymbol{\theta}) = \mathrm{Cov}[\boldsymbol{s}(\boldsymbol{\theta})\boldsymbol{s}(\boldsymbol{\theta})^\top] = \mathbb{E}_{\boldsymbol{x}\sim p(\boldsymbol{x}|\boldsymbol{\theta})}[\nabla_{\boldsymbol{\theta}} \log p(\boldsymbol{x}|\boldsymbol{\theta}) \nabla_{\boldsymbol{\theta}} \log p(\boldsymbol{x}|\boldsymbol{\theta})^\top]$$

And here comes the theorem.

Theorem 3.2.2.1 (Equivalence Between FIM and Expectation of Hessian of NLL)

Let $\log p(\boldsymbol{x}|\boldsymbol{\theta})$ be twice-differentiable, and ceratin conditions are satisfied. Then, the Fisher information matrix equals the expectation of Hessian of the negative log likelihood, i.e.,

$$\mathbb{E}_{\mathcal{D}\sim p(\mathcal{D}|\boldsymbol{\theta})}[\mathbf{H}(\mathcal{D})] = N\mathbf{F}(\boldsymbol{\theta})$$

where $\mathbf{H}$ is Hessian of N iid samples, $\mathcal{D}$, and $\mathbf{F}(\boldsymbol{\theta})$ is the Fisher information matrix.

Proof. Search for internet. □

3.3 Conjugate Priors

Calculating the posterior distribution can be untractable, if the prior and likelihood does not match well. Here the well-matching prior according to the likelihood is given, called the **conjugate priors**.

likelihood	unknown parameters	conjugate prior	parameters(prior)	posterior

3.4 Uninformative Priors

3.5 Hierarchical Priors

3.6 Empirical Priors

Chapter 4

Decision Theory

After

Chapter 5

Graphical Models

Chapter 6

Information Theory

Chapter 7

Optimization

```
# sample code block
def is_prime(n):
    if n < 2:
        return false
    for i in range(n**0.5, 2):
        if n % i == 0
            return false

    return true
```